

WORKSHOP À NANTES : ANALYSE SEMICLASSIQUE LE LONG DE LA LOIRE

SCHEDULE

	Tuesday 25/08	Wednesday 26/08	Thursday 27/08
09 :30–10 :30		Averseng	Grugier
10 :30–11 :00		Coffee	Coffee
11 :00–12 :00		Curely	Jézéquel
12 :00–14 :00		Lunch	Lunch
14 :00–14 :30	Arrival / Coffee	Coffee	Coffee
14 :30–15 :30	Ingremeau	Lechaux	
15 :30–16 :00	Coffee	Coffee	
16 :00–17 :00	Verdasco Ramos	Sabin	
19 :00		Dinner	

PROGRAM

Martin Averseng. Localization of the eigenfunctions of the Bloch-Torrey operator.

Abstract : The Bloch-Torrey operator, $L(h) = h^2\Delta + iV(x)$ (say, over a bounded domain with Dirichlet conditions, V having no critical points) is a non-selfadjoint operator appearing in physical contexts such as superconductivity and nuclear magnetic resonance imaging. The spectrum of the operator in the limit $h \rightarrow 0$ has been investigated in several works by the semiclassical analysis community over the past 20 years. In many of these, quasi-modes for $L(h)$ are constructed, all exhibiting characteristic localization properties : their mass concentrates near points where $\text{grad } V$ is orthogonal to the boundary, as $h \rightarrow 0$. It is strongly suspected that these quasi-modes are close to true eigenfunctions, but not yet proven. In this talk, I will present a result obtained in collaboration with Nicolas Frantz, Frédéric Hérau and Nicolas

Raymond, which establishes the expected localization property of eigenfunctions for a flattened version of the Bloch-Torrey operator near one of these special boundary points—we believe that this result is key to answering the original question for the curved problem. A sketch of the proof will be presented, with the following two main ideas : (i) the classical method of Agmon estimates, to treat a 1-dimensional problem resulting from an adiabatic projection and (ii) an argument using semiclassical pseudodifferential operators with operator-valued symbols, to control the non-adiabatic remainder.

Marianne Curely. Propagation of wave packets close to conical intersections.

Abstract : In this presentation, we study the propagation of wave packets close to conical intersections with respect to a system of two Schrödinger equations presenting a codimension 2 crossing. We focus on the dynamics that occur when the wave packets pass through an area close to the crossing and our main results provide an explicit formula for the outgoing wave packet, with a complete description of its phase and of the classical trajectories it follows, including a drift.

Emma Grugier. Small eigenvalues of non-reversible metastable diffusion processes with Neumann boundary conditions.

Abstract : The configuration of a molecule is determined by two main factors : its potential energy and its thermal agitation. This behavior can be described by a stochastic differential equation in which thermal agitation is represented by Brownian motion. Spectral analysis of the diffusion operator associated with the process allows us to characterize the long-time behavior of the molecule at low temperatures.

Maxime Ingremeau. Multifractality of eigenfunctions on quantum star graphs.

Abstract : Multifractality is a property of semiclassical eigenfunctions that are neither fully localised nor delocalised. After reviewing the existing definitions and mathematical results concerning multifractality, we will present new families of multifractal eigenfunctions on large quantum star graphs, and classify their possible asymptotic behaviour. This talk is based on the results obtained by my PhD student Marius Nietschmann.

Malo Jézéquel. Internal waves and viscosity.

Abstract : In this talk, I will discuss the geometric focusing of internal waves in the presence of viscosity. In particular, I will explain how one can understand the low viscosity regime by replacing the original domain on which the problem is set (a bounded domain in $\mathbb{R}^2$) by another surface in $\mathbb{C}^2$. Joint work with Jian Wang.

Julien Lechaux. Dynamics of the Magnetic Schrödinger Equation.

Abstract : In this talk, I will discuss recent results on the long-time dynamics of the magnetic Schrödinger equation on the two-dimensional flat torus. Although the underlying classical dynamics is completely integrable, the presence of the magnetic field gives rise to regularization and equidistribution effects.

Julien Sabin. A many body version of gaussian beams and zonal functions.

Abstract : I will present a construction of orthonormal families of quasimodes of the laplacian on a compact manifold that generalize the concentration displayed by gaussian beams (around a geodesic) or zonal functions (around a point). In the same way that gaussian beams and zonal functions have an optimal growth of L^p norms, these orthonormal families saturate a 'many-body' version of these L^p bounds. This is a joint work with Rupert Frank (Munich).

Santiago Verdasco Ramos. Semiclassical measures of point-perturbations of the Laplacian.

Abstract : Point-perturbations of the Laplacian are a type of singular perturbations of the Laplacian of particular interest to the quantum chaos community in Mathematical Physics. This model was proposed to understand the transition between chaotic and integrable quantum dynamics. In this talk, we will explore this model from the Correspondence Principle, showing that on a very generic setting, high-energy eigenfunctions have invariant semiclassical measure, but on the spheres, this invariance may be broken.